



Master Presenter's Manual

for

Data Structures in C

Curriculum Code: OV-1504

***Effective from: February 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	February 2024	New	-

Table of Contents

S#.	Details	Page No.
1.	Introduction	1
2.	Information on Module Allocation	2
3.	Week-wise Session Schedule	3
4.	Session Coverage	4
5.	Deliverables	8
6.	Library References	8
7.	Evaluation Strategy	9
8.	Classroom and Lab Pre-requisites	10

1. Introduction

At the end of this course, students will be able to:

- Describe data structures and types of data structures
- Write a basic C program that uses binary search algorithm
- Explain the Tree data structure
- Use Arrays, Pointers, and Structures in a program
- Implement sorting algorithms in 'C'
- Describe how to implement linked lists using 'C'
- Implement a stack in a 'C' program
- Implement queues in C using arrays and linked lists
- Describe binary trees
- Explain how to build and initialize matrices
- Implement hash tables in C
- Describe how to implement graphs using 'C'

2. Information on Module Allocation

Module	Online (No. of Hours)	Skill Code
Data Structures in C	24	75

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 DS - TL1	Session 2 DS - TL2	Session 3 DS - TL3	Session 4 DS - TL4
2	Session 5 DS - TL5	Session 6 DS - TL6	Session 7 DS - TL7	Session 8 DS - TL8
3	Session 9 DS - TL9	Session 10 DS - TL10	Session 11 DS - TL11	Session 12 DS - TL12

DS: Data Structures in C

TL: Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	DS – TL1	All the topics as listed below from Session 1 of Data Structures in C book should be covered in this session. <u>Session 1 – Introduction to C</u> ➤ Write a basic C program ➤ Use Conditions and Loops in a program ➤ Use Functions in a program ➤ Use Arrays, Pointers, and Structures in a program	Data Structures in C SG – Session 1 XP – Session 1 TG – Session 1
2	DS – TL2	All the topics as listed below from Session 2 of Data Structures in C book should be covered in this session. <u>Session 2 – Introduction to Data Structures and Algorithms</u> ➤ Describe data structures and types of data structures ➤ Define algorithms ➤ Explain sequential search and binary search algorithms ➤ Write a basic C program that uses sequential search algorithm ➤ Write a basic C program that uses binary search algorithm ➤ Explain the Tree data structure	Data Structures in C SG – Session 2 XP – Session 2 TG – Session 2
3	DS – TL3	All the topics as listed below from Session 3 of Data Structures in C book should be covered in this session. <u>Session 3 – Sorting Algorithms</u> ➤ Describe sorting algorithms and its types ➤ Explain the use of various types of sorting algorithms ➤ Implement sorting algorithms in 'C'	Data Structures in C SG – Session 3 XP – Session 3 TG – Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
4	DS – TL4	The students should practice all the programs given in Session 1 to Session 3 of <i>Data Structures in C</i> book.	Data Structures in C SG – Session 1-3
5	DS – TL5	All the topics as listed below from Session 4 of <i>Data Structures in C</i> book should be covered in this session. <u>Session 4 – Linked Lists</u> ➤ Define linked lists ➤ State the advantages of linked lists ➤ Differentiate between linked lists and array ➤ Explain the application scenarios of linked lists ➤ Explain Singly-linked list, Doubly-linked list, and Circular linked list ➤ Describe how to implement linked lists using 'C'	Data Structures in C SG – Session 4 XP – Session 4 TG – Session 4
6	DS – TL6	All the topics as listed below from Session 5 of <i>Data Structures in C</i> book should be covered in this session. <u>Session 5 – Stacks</u> ➤ Explain stack ➤ Describe the operations of stack ➤ Explain the implementation of stack using an array and linked list ➤ Implement a stack in a 'C' program	Data Structures in C SG – Session 5 XP – Session 5 TG – Session 5
7	DS – TL7	All the topics as listed below from Session 6 of <i>Data Structures in C</i> book should be covered in this session. <u>Session 6 – Queues</u> ➤ Define queues ➤ Describe the features of queues ➤ Explain the operations on	Data Structures in C SG – Session 6 XP – Session 6 TG – Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		queues ➤ Explain the functions used to implement queues ➤ Implement queues in C using arrays and linked lists	
8	DS – TL8	The students should practice all the programs given in Session 4 to Session 6 of <i>Data Structures in C</i> book.	Data Structures in C SG – Session 4-6
9	DS – TL9	All the topics as listed below from Session 7 of <i>Data Structures in C</i> book should be covered in this session. <u>Session 7 – Binary Trees</u> ➤ Describe binary trees ➤ Explain traversals in binary tree ➤ Describe binary search trees ➤ Explain various operations on a binary search tree ➤ Describe threaded binary tree ➤ Explain height balanced tree	Data Structures in C SG – Session 7 XP – Session 7 TG – Session 7
10	DS – TL10	All the topics as listed below from Session 8 of <i>Data Structures in C</i> book should be covered in this session. <u>Session 8 – Matrices and Hash</u> ➤ Define matrices ➤ Explain how to build and initialize matrices ➤ Explain manipulation of matrices ➤ Define hash tables ➤ Learn how to build hash tables ➤ Describe collision management ➤ Implement hash tables in C	Data Structures in C SG – Session 8 XP – Session 8 TG – Session 8
11	DS – TL11	All the topics as listed below from Session 9 of <i>Data Structures in C</i> book should be covered in this session. <u>Session 9 – Graphs</u> ➤ Define graphs	Data Structures in C SG – Session 9 XP – Session 9 TG – Session 9

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Describe the need for graphs ➤ Explain how to build a graph ➤ Explain the shortest path to traverse in a graph ➤ Describe how to implement graphs using 'C'	
12	DS – TL12	The students should practice all the programs given in Session 7 to Session 9 of <i>Data Structures in C</i> book.	Data Structures in C SG – Session 7-9

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Student:

Module	Deliverable
Data Structures in C	Kit Code: OV-1504-KIT01-01-A Kit Name: Kit of 1 title for OV-1504 Data Structures C

Deliverables for Faculty:

Module	Deliverable
Data Structures in C	Kit Code: OV-1504-KIT01-01-A Kit Name: Kit of 1 title for OV-1504 Data Structures C
Course	Classroom Presentations Trainer Guide Master PM

6. Library References

➤ Data Structures in C by Noel Kalicharan
➤ Data Structure Through C by Yashavant P. Kanetkar

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for PASS

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for CREDIT and

Overall Weighted Marks $\geq 75\%$ qualifies for DISTINCTION

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination (E1)

Marks: 50

Duration: 60 minutes

Type: Objective

Sr. No.	Name of the Subject	Distribution of Marks
1.	Data Structures in C	50

Final Examination (R1)

Marks: 50

Duration: 120 minutes

Type: Practical

Sr. No.	Name of the Subject	Distribution of Marks
1.	Data Structures in C	50

Evaluation Details

Practical – 50%

OES – 50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the module of this course have been described as follows:

8.1 Software Requirements

Microsoft Windows 10 or higher
NetBeans IDE latest version

8.2 Recommended Hardware

Intel Core i3/i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

~~ End of Document ~~